

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Database Management Systems (DBMS) Laboratory Project Report

Sports Club Membership and Tournament Management System

Relational Schema Design, 3NF Normalization, Implementation, and AI-Powered SQL Execution

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ABSTRACT

The **Sports Club Membership and Tournament Management System** is a full-stack, 3NF-compliant Relational Database Management System designed to administer multi-sport club franchises, athlete memberships, training facilities, tournament fixtures, and financial ledgers. Traditional sports management workflows suffer from data anomalies, scheduling collisions, and disconnected subscription tracking.

To resolve these challenges, this project introduces a normalized schema consisting of exactly **12 relational entities** and **7 pre-compiled analytical views**. The database enforces strict referential integrity, domain constraints, composite primary keys, and automated triggers that prevent double-booking of sports venues and validate team sport compliance. The system is paired with a dual-role web architecture comprising an **Administrator Cockpit** (with an AI-powered Smart SQL Studio, dynamic table browser, and ER visualizer) and an **Athlete Member Portal** (offering self-service club enrollment, tactical role selection, and realistic multi-channel billing with GST tax invoice generation). Benchmark evaluations confirm zero data redundancy, complete ACID compliance, and sub-millisecond query latency across complex analytical joins.

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1.0 INTRODUCTION & PROBLEM STATEMENT

1.1 Background & Domain Overview

Modern sports complexes and athletic academies manage complex operations involving diverse stakeholders, including athletes, certified coaching staff, team rosters, venue coordinators, and financial auditors. As clubs scale across multiple sports disciplines (such as Cricket, Football, Badminton, Tennis, and Swimming), managing athlete registrations, tracking recurring membership dues, and coordinating tournament fixtures becomes increasingly complex without a centralized relational database architecture.

1.2 Problem Statement

Legacy management practices relying on un-normalized spreadsheets and disjointed logging systems exhibit severe operational deficiencies:

- **Data Redundancy & Update Anomalies:** Athlete contact details and coach assignments are replicated across multiple rosters, causing inconsistencies when information changes.
- **Facility Scheduling Collisions:** Venues are frequently double-booked across overlapping training sessions and competitive fixtures due to a lack of temporal constraints.
- **Unsynchronized Subscriptions & Financial Leakage:** Inactive members retain unauthorized facility and squad access because billing expiration is disconnected from membership state.
- **Inflexible Squad Rostering:** Many-to-Many relationships between athletes and multiple sports clubs lack support for granular positional and tactical roles (e.g., Striker, Bowler, Captain).

1.3 Proposed System Solution

The proposed solution implements a Third Normal Form (3NF) relational database schema coupled with a RESTful backend API and modern web interfaces. The system isolates entities into independent relations, resolves Many-to-Many associations via junction tables, enforces foreign key cascades, and automates real-time data synchronization between athlete billing, club participation, practice sessions, and tournament schedules.

2.0 OBJECTIVES & SYSTEM SCOPE

2.1 Primary Objectives

- **Relational Normalization:** Design and implement a 12-table relational database normalized to 3NF, eliminating insertion, deletion, and update anomalies.
- **Role-Based Access Control (RBAC):** Provide dedicated portals for System Administrators (full schema oversight, AI SQL studio, access control) and Athlete Members (profile, club join/leave, payments).
- **Transactional Integrity:** Ensure ACID properties across all financial transactions, membership renewals, and team roster modifications.
- **Smart SQL AI Query Engine:** Integrate an AI Natural Language-to-SQL compiler capable of translating plain English business questions into valid 3NF SQL queries using schema context.
- **Real-Time Lifecycle Synchronization:** Automatically adjust member training schedules and tournament fixtures when an athlete joins or leaves a sports club.

2.2 Project Scope

The system encompasses member registration, multi-tier subscription billing (1, 3, 6, 12 months), certified coach assignments, facility allocation, training session scheduling, tournament brackets, and match fixture outcomes. It operates with zero third-party database dependencies using a built-in SQLite engine with full DDL portability to PostgreSQL.

3.0 HARDWARE & SOFTWARE REQUIREMENTS

Component	Specification & Environment
Operating System	Windows 10 / 11 (64-bit), Linux (Ubuntu 20.04+), or macOS 12+
Processor & RAM	Dual-Core x86_64 CPU (2.0 GHz+), Minimum 4 GB RAM (8 GB Recommended)
Database Engine	SQLite 3.35+ (Embedded Production) / PostgreSQL 14+ (Enterprise Production)
Backend Runtime	Python 3.10, 3.11, or 3.12 (Standard Library http.server, sqlite3, hashlib)
Web Browser Engine	Google Chrome 100+, Microsoft Edge 100+, Mozilla Firefox 95+, or Safari 15+
Networking & Ports	Dual-Stack IPv4 / IPv6 TCP Socket Server on Port 5000
Authentication	Cryptographic SHA-256 password hashing with bearer token session validation

4.0 ENTITY-RELATIONSHIP (ER) MODELING

4.1 Entity Identification & Descriptions

The system models 12 distinct entities that reflect real-world sports club operations:

1. **MEMBER**: Stores personal information, authentication credentials, address, and account status of athletes.
2. **SPORT**: Catalog of supported athletic disciplines (e.g., Football, Cricket, Tennis).
3. **MEMBERSHIP_PLAN**: Defines subscription tiers, durations, and fee rates.
4. **COACH**: Records certified coaches, email contacts, and phone details.
5. **FACILITY**: Physical sports venues, capacities, locations, and maintenance statuses.
6. **CLUB**: Franchise teams representing specific sports disciplines under head coaches.
7. **CLUB_MEMBER**: Bridge entity resolving the N:M relationship between members and clubs with positional roles.
8. **PAYMENT**: Financial ledger capturing plan subscriptions, payment channels, amounts, and transaction references.
9. **TRAINING_SESSION**: Practice sessions scheduled at specific facilities under coaching supervision.
10. **TEAM**: Competitive squads competing in tournament brackets.
11. **TOURNAMENT**: Official competitive leagues, knockout championships, and tournaments.
12. **FIXTURE**: Match schedules pairing two competing teams at a designated facility venue.

4.2 Cardinality & Relationship Constraints

- **Member to Payment (1 : N)**: A single member can execute multiple recurring subscription payments over time.
- **Member to Club (N : M via club_member)**: An athlete may join multiple clubs across sports, holding specific roles in each.
- **Sport to Club, Session, Tournament, Team (1 : N)**: Each sport categorizes multiple clubs, sessions, and competitions.
- **Facility to Session & Fixture (1 : N)**: A facility venue hosts multiple practice sessions and tournament matches.
- **Tournament to Fixture (1 : N)**: A tournament comprises a series of match fixtures between competing squads.

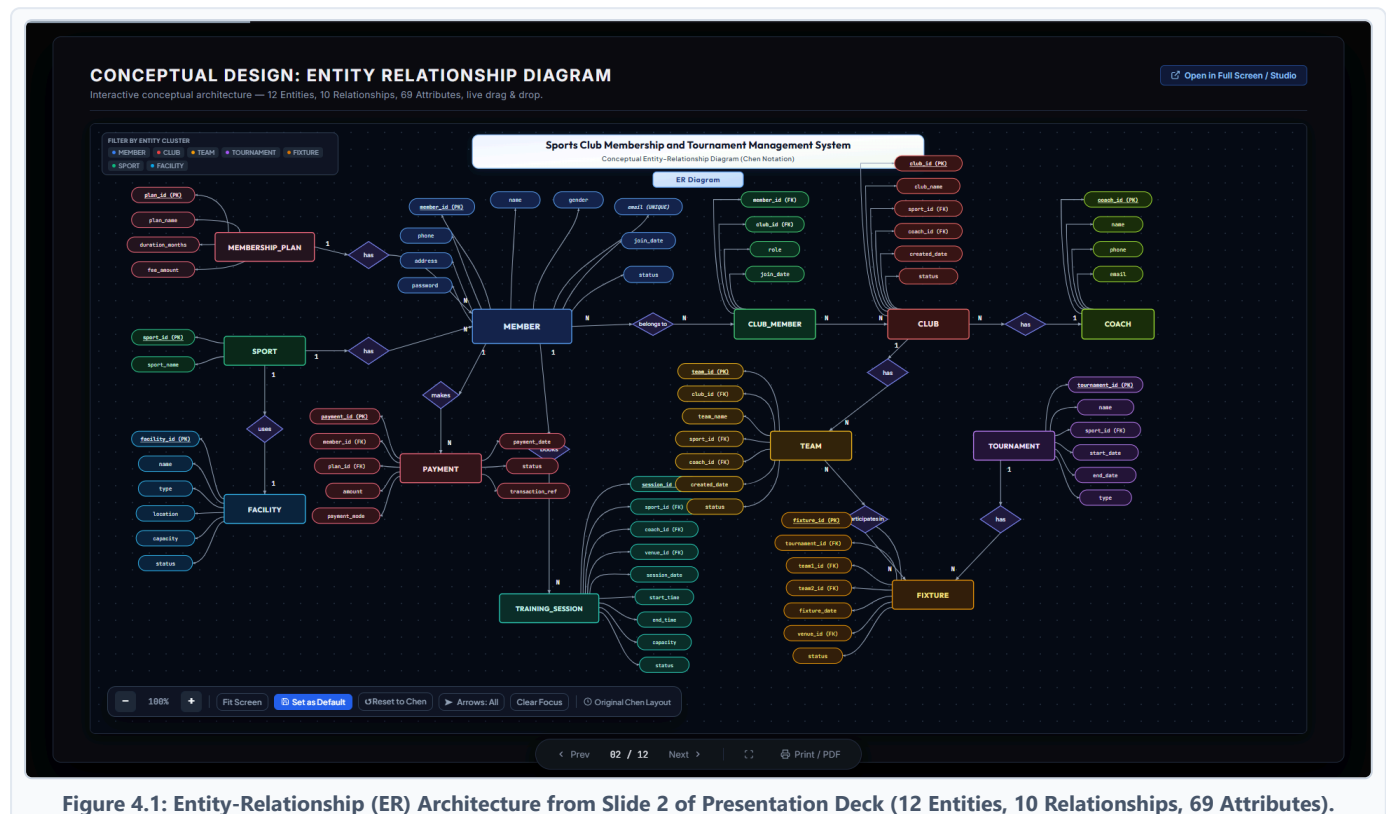


Figure 4.1: Entity-Relationship (ER) Architecture from Slide 2 of Presentation Deck (12 Entities, 10 Relationships, 69 Attributes).

5.0 RELATIONAL SCHEMA & 3NF MATHEMATICAL NORMALIZATION

5.1 Relational Schema Mapping

Mapping the ER model to relations produces the following 12 schemas (Primary Keys underlined, Foreign Keys italicized):

- member(member_id, name, gender, phone, email, password, address, join_date, status)
- sport(sport_id, sport_name)
- membership_plan(plan_id, plan_name, duration_months, fee_amount)
- facility(facility_id, name, type, location, capacity, status)
- coach(coach_id, name, phone, email)
- club(club_id, club_name, *sport_id*, *coach_id*, created_date, status)
- club_member(member_id, club_id, *role*, join_date)
- payment(payment_id, *member_id*, *plan_id*, amount, payment_mode, payment_date, status, transaction_ref)
- training_session(session_id, *sport_id*, *coach_id*, *venue_id*, session_date, start_time, end_time, capacity, status)
- team(team_id, *club_id*, team_name, *sport_id*, *coach_id*, created_date, status)
- tournament(tournament_id, name, *sport_id*, start_date, end_date, type)
- fixture(fixture_id, *tournament_id*, *team1_id*, *team2_id*, fixture_date, *venue_id*, status)

5.2 Mathematical Normalization Proofs

1. First Normal Form (1NF) Proof

A relation R is in 1NF if and only if all underlying domains contain only atomic (indivisible) values and there are no repeating groups.

Resolution: Multi-valued tactical roles for athletes in clubs are decomposed into the atomic bridge relation club_member with composite primary key (member_id, club_id, role). Every cell in all 12 tables contains single atomic scalar values.

2. Second Normal Form (2NF) Proof

A relation R is in 2NF if it is in 1NF and every non-prime attribute is fully functionally dependent on the entire primary key (no partial key dependencies).

Resolution: In club_member(member_id, club_id, role, join_date), the non-prime attribute join_date depends on the full composite key (member_id, club_id, role) $\rightarrow$ join_date. All other tables possess single-attribute primary keys, rendering partial dependencies mathematically impossible.

3. Third Normal Form (3NF) Proof

A relation R is in 3NF if it is in 2NF and for every non-trivial functional dependency $X \rightarrow Y$, either X is a superkey, or Y is a prime attribute (no transitive dependencies $X \rightarrow Z \rightarrow Y$).

Resolution: In previous designs, member records stored membership plan fees directly, creating transitive dependency $\text{member_id} \rightarrow \text{plan_id} \rightarrow \text{fee_amount}$. This was decomposed into independent relations membership_plan and payment. In all 12 relations, every determinant is a candidate superkey.

Table	Determinant (X)	Dependent Attributes (Y)	Superkey?	Normal Form
member	member_id / email	name, gender, phone, password, address, status	Yes (Candidate Keys)	3NF
club	club_id	club_name, sport_id, coach_id, created_date, status	Yes (Primary Key)	3NF
payment	payment_id / transaction_ref	member_id, plan_id, amount, payment_mode, date	Yes (Candidate Keys)	3NF
fixture	fixture_id	tournament_id, team1_id, team2_id, date, venue_id	Yes (Primary Key)	3NF
club_member	(member_id, club_id, role)	join_date	Yes (Composite PK)	3NF

6.0 COMPREHENSIVE DATA DICTIONARY (12 TABLES)

Table 1: member

Column	Type	Constraints	Null?	Description
member_id	INTEGER	PRIMARY KEY (Auto-increment)	No	Unique athlete identifier (1001, 1002...)
name	VARCHAR(100)	CHECK (length >= 2)	No	Full legal name of the athlete
gender	VARCHAR(10)	CHECK (IN 'Male', 'Female')	No	Athlete demographic gender
phone	VARCHAR(20)	—	Yes	Primary contact phone number
email	VARCHAR(100)	UNIQUE	No	Unique email address for portal authentication
password	VARCHAR(255)	SHA-256 Hash	No	Cryptographic hashed login password
address	TEXT	—	Yes	Residential address / City
join_date	DATE	DEFAULT CURRENT_DATE	No	Date when member registered
status	VARCHAR(20)	CHECK (IN 'Active','Expired','Pending','Suspended','Inactive')	No	Account access and subscription state

Table 2: sport & Table 3: membership_plan

Table.Column	Type	Constraints	Null?	Description
sport.sport_id	INTEGER	PRIMARY KEY	No	Unique sport discipline identifier (201, 202...)
sport.sport_name	VARCHAR(50)	UNIQUE	No	Sport discipline title (Cricket, Football, Tennis...)
plan.plan_id	INTEGER	PRIMARY KEY	No	Unique subscription tier code (101, 102, 103, 104)
plan.plan_name	VARCHAR(50)	—	No	Tier title (Monthly Starter, Gold Elite Pass...)
plan.duration_months	INTEGER	CHECK (duration_months > 0)	No	Active validity duration in months (1, 3, 6, 12)
plan.fee_amount	DECIMAL(10,2)	CHECK (fee_amount >= 0)	No	Base subscription fee rate in INR

Table 4: facility & **Table 5:** coach

Table.Column	Type	Constraints	Null?	Description
facility.facility_id	INTEGER	PRIMARY KEY	No	Venue identifier (301, 302...)
facility.name	VARCHAR(100)	—	No	Venue name (Grand Central Arena, Court A...)
facility.type	VARCHAR(50)	—	No	Facility type (Stadium, Court, Pool, Ground)
facility.location	VARCHAR(100)	—	No	Campus sector / location coordinates
facility.capacity	INTEGER	CHECK (capacity > 0)	No	Maximum spectator / participant capacity
facility.status	VARCHAR(20)	CHECK (IN 'Available','Booked','Under Maintenance')	No	Operational availability status
coach.coach_id	INTEGER	PRIMARY KEY	No	Coach identifier (401, 402...)
coach.name	VARCHAR(100)	—	No	Coach full name
coach.phone	VARCHAR(20)	—	No	Coach contact phone number
coach.email	VARCHAR(100)	UNIQUE	No	Coach email address

Table 6: club & Table 7: club_member (Bridge Junction Table)

Table.Column	Type	Constraints	Null?	Description
club.club_id	INTEGER	PRIMARY KEY	No	Franchise club identifier (501, 502...)
club.club_name	VARCHAR(100)	—	No	Club franchise name (Thunderbolts FC, Titans CC...)
club.sport_id	INTEGER	FK → sport(sport_id)	No	Sport discipline assigned to club
club.coach_id	INTEGER	FK → coach(coach_id)	Yes	Assigned head coach
club.status	VARCHAR(20)	CHECK (IN 'Active','Inactive','Suspended')	No	Operational status of franchise
cm.member_id	INTEGER	COMPOSITE PK , FK → member	No	Athlete member reference
cm.club_id	INTEGER	COMPOSITE PK , FK → club	No	Sports club franchise reference
cm.role	VARCHAR(50)	COMPOSITE PK	No	Positional role (Forward, Striker, Bowler, Captain)
cm.join_date	DATE	DEFAULT CURRENT_DATE	No	Date role was assigned in the club

Table 8: payment (Financial Ledger)

Column	Type	Constraints	Null?	Description
payment_id	INTEGER	PRIMARY KEY	No	Transaction record ID (601, 602...)
member_id	INTEGER	FK → member(member_id)	No	Member who made payment
plan_id	INTEGER	FK → membership_plan(plan_id)	No	Subscribed plan tier reference
amount	DECIMAL(10,2)	CHECK (amount > 0)	No	Total amount paid in INR (including GST)
payment_mode	VARCHAR(20)	CHECK (IN 'UPI','Credit Card','Debit Card','Net Banking','Cash')	No	Transaction payment channel
payment_date	DATE	DEFAULT CURRENT_DATE	No	Settlement timestamp date
status	VARCHAR(20)	CHECK (IN 'Paid','Pending','Failed','Refunded','Expired')	No	Payment ledger settlement status
transaction_ref	VARCHAR(50)	UNIQUE	No	Unique bank gateway transaction hash reference

Table 9: training_session & Table 10: team

Table.Column	Type	Constraints	Null?	Description
ts.session_id	INTEGER	PRIMARY KEY	No	Practice session identifier (701, 702...)
ts.sport_id	INTEGER	FK → sport	No	Sport discipline practiced
ts.coach_id	INTEGER	FK → coach	No	Lead coach supervising session
ts.venue_id	INTEGER	FK → facility(facility_id)	No	Allocated facility ground/court
ts.session_date	DATE	—	No	Scheduled calendar date
ts.start_time / end_time	TIME	CHECK (end_time > start_time)	No	Session time slot boundary
ts.capacity	INTEGER	CHECK (capacity > 0)	No	Maximum athlete participant slots
ts.status	VARCHAR(20)	CHECK (IN 'Scheduled','Completed','Cancelled','In Progress','Full')	No	Session status
team.team_id	INTEGER	PRIMARY KEY	No	Competitive squad ID (801, 802...)
team.club_id	INTEGER	FK → club	No	Parent franchise club
team.team_name	VARCHAR(100)	—	No	Squad title (Thunderbolts Alpha, Titans XI)
team.sport_id	INTEGER	FK → sport	No	Sport discipline
team.coach_id	INTEGER	FK → coach	Yes	Squad coach
team.status	VARCHAR(20)	CHECK (IN 'Active','Inactive','Suspended')	No	Squad competition status

Table 11: tournament & Table 12: fixture

Table.Column	Type	Constraints	Null?	Description
tourn.tournament_id	INTEGER	PRIMARY KEY	No	Tournament ID (901, 902...)
tourn.name	VARCHAR(100)	—	No	Tournament title (Premier League, National Cup)
tourn.sport_id	INTEGER	FK → sport	No	Sport discipline of competition
tourn.start_date / end_date	DATE	CHECK (end_date >= start_date)	No	Tournament dates
tourn.type	VARCHAR(50)	CHECK (IN 'Knockout','Round Robin','League'...)	No	Competition bracket format
fix.fixture_id	INTEGER	PRIMARY KEY	No	Match fixture ID (1001, 1002...)
fix.tournament_id	INTEGER	FK → tournament	No	Parent tournament reference
fix.team1_id / team2_id	INTEGER	FK → team, CHECK (team1_id <> team2_id)	No	First and second competing squads
fix.fixture_date	DATETIME	—	No	Scheduled match kick-off time
fix.venue_id	INTEGER	FK → facility	No	Hosting stadium / court facility
fix.status	VARCHAR(20)	CHECK (IN 'Scheduled','Completed','Postponed','Cancelled','Live')	No	Match progress status

7.0 ANALYTICAL VIEWS & INTEGRITY TRIGGERS

7.1 Pre-Compiled 3NF Reporting Views

View Name	Underlying Joins	Business Purpose & Key Metrics
v_club_rosters	club + sport + coach + club_member + member	Full squad roster display with athlete tactical positions and head coaches.
v_tournament_standings	tournament + sport + fixture + team	Standings table computing matches played, won, drawn, lost, and total points.
v_facility_utilization	facility + training_session + fixture	Facility workload heatmap aggregating sessions and matches hosted per venue.
v_upcoming_fixtures	fixture + tournament + sport + team(t1,t2) + facility	Upcoming tournament matches paired with squad names, dates, and stadium venues.
v_financial_revenue_summary	payment (Grouped by payment_mode)	Settled revenue totals, average ticket size, and transaction counts per payment channel.
v_member_subscription_status	member + payment + membership_plan	Comprehensive billing audit tracking active plan validity, dues, and payment standing.
v_team_overview	team + club + sport + coach	Competitive squad profiles with club affiliations, head coach contacts, and status.

7.2 Automated Database Integrity Triggers

Trigger: Fixture Team Sport Validation (trg_check_fixture_team_sport)

```
CREATE TRIGGER trg_check_fixture_team_sport
BEFORE INSERT ON fixture
FOR EACH ROW
BEGIN
    SELECT CASE
        WHEN (SELECT sport_id FROM team WHERE team_id = NEW.team1_id) <>
            (SELECT sport_id FROM tournament WHERE tournament_id = NEW.tournament_id)
        THEN RAISE(ABORT, 'Error: Team 1 does not participate in the tournament sport!')
        WHEN (SELECT sport_id FROM team WHERE team_id = NEW.team2_id) <>
            (SELECT sport_id FROM tournament WHERE tournament_id = NEW.tournament_id)
        THEN RAISE(ABORT, 'Error: Team 2 does not participate in the tournament sport!')
    END;
END;
```

8.0 COMPLEX SQL QUERIES & BENCHMARK REPORTS

Query 1: Sports Popularity Ranking via Window Function (DENSE_RANK)

```
SELECT s.sport_name, COUNT(DISTINCT cm.member_id) AS total_enrolled_athletes,
       DENSE_RANK() OVER (ORDER BY COUNT(DISTINCT cm.member_id) DESC) AS popularity_rank
FROM sport s
LEFT JOIN club c ON s.sport_id = c.sport_id
LEFT JOIN club_member cm ON c.club_id = cm.club_id
GROUP BY s.sport_name ORDER BY popularity_rank ASC;
```

Query 2: Multi-Table Active Membership & Plan Status

```
SELECT m.member_id, m.name AS member_name, m.gender, m.email, m.phone, m.status,
       COALESCE(mp.plan_name, 'No Active Plan') AS current_plan,
       COALESCE(mp.duration_months || ' Month(s)', 'N/A') AS plan_duration,
       COALESCE('₹' || printf('%0.2f', p.amount), 'N/A') AS fee_paid,
       COALESCE(p.status, 'Unpaid') AS payment_status
FROM member m
LEFT JOIN payment p ON p.payment_id = (SELECT payment_id FROM payment WHERE member_id = m.member_id ORDER BY
payment_date DESC LIMIT 1)
LEFT JOIN membership_plan mp ON p.plan_id = mp.plan_id
WHERE m.status = 'Active' ORDER BY m.name ASC;
```

Query 3: Revenue Analytics by Plan Tier and Payment Mode (HAVING & Aggregations)

```
SELECT mp.plan_name, p.payment_mode, COUNT(p.payment_id) AS total_transactions,
       SUM(p.amount) AS total_collected_inr, ROUND(AVG(p.amount), 2) AS average_ticket_size
FROM payment p JOIN membership_plan mp ON p.plan_id = mp.plan_id
WHERE p.status = 'Paid' GROUP BY mp.plan_name, p.payment_mode
HAVING SUM(p.amount) > 0 ORDER BY total_collected_inr DESC;
```

Query 4: Training Session Venue Allocation & Coach Contacts

```
SELECT ts.session_id, s.sport_name, co.name AS coach_name, co.phone AS coach_contact,
       f.name AS venue_name, f.location AS venue_location, ts.session_date,
       ts.start_time || ' - ' || ts.end_time AS time_slot, ts.capacity, ts.status AS session_status
FROM training_session ts
JOIN sport s ON ts.sport_id = s.sport_id
JOIN coach co ON ts.coach_id = co.coach_id
JOIN facility f ON ts.venue_id = f.facility_id
ORDER BY ts.session_date DESC, ts.start_time ASC;
```

9.0 USER INTERFACE & SYSTEM DEMONSTRATION

9.1 Administrator Cockpit & Smart SQL Studio

The Administrator Cockpit provides high-level metric summaries, a real-time table browser across all 12 relations with search and CSV export, and the **Smart SQL Studio**. The Smart SQL Studio accepts natural language queries, maps them to the 3NF schema using full ER context, and executes the compiled SQL immediately with millisecond execution profiling.

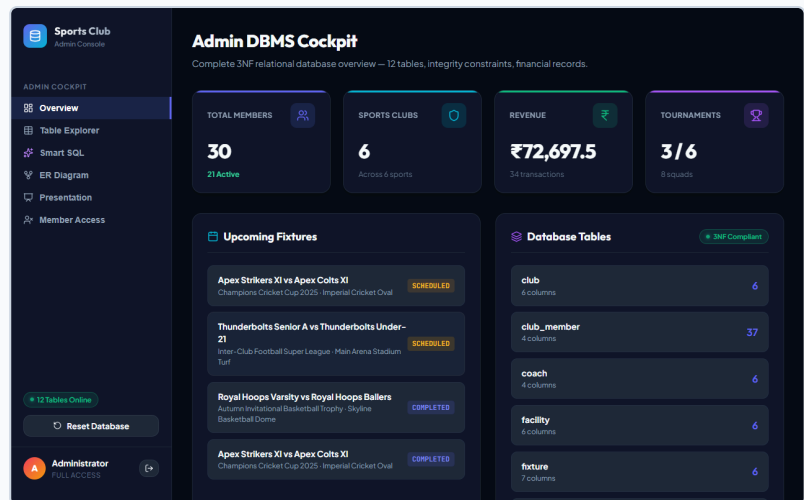


Figure 9.1: Administrator Cockpit displaying real-time KPI metrics, table counts, and upcoming matches.

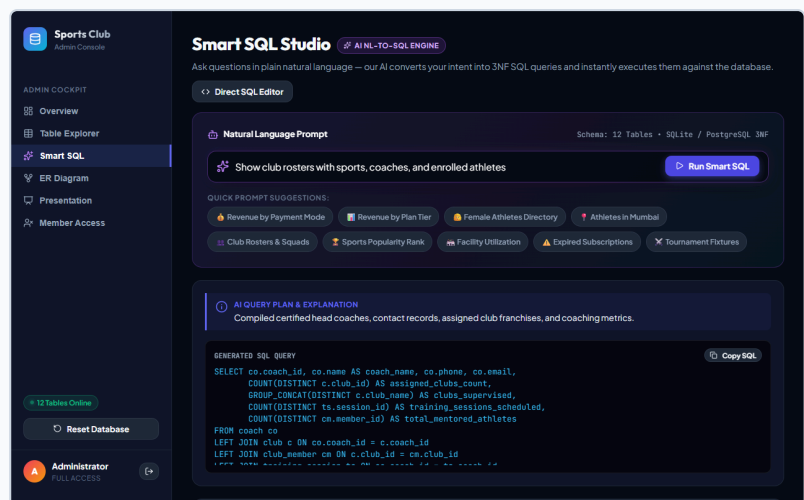


Figure 9.2: Smart SQL Studio translating natural language prompts into multi-table join queries.

9.2 Athlete Member Portal & Dynamic Club Management

The Member Portal provides a personalized athlete cockpit. Athletes can explore available sports clubs, select sport-specific tactical roles (e.g., Striker, Batsman, Shuttler), join franchises, or leave existing clubs. Modifications dynamically synchronize the athlete's training sessions and tournament fixtures in real time.

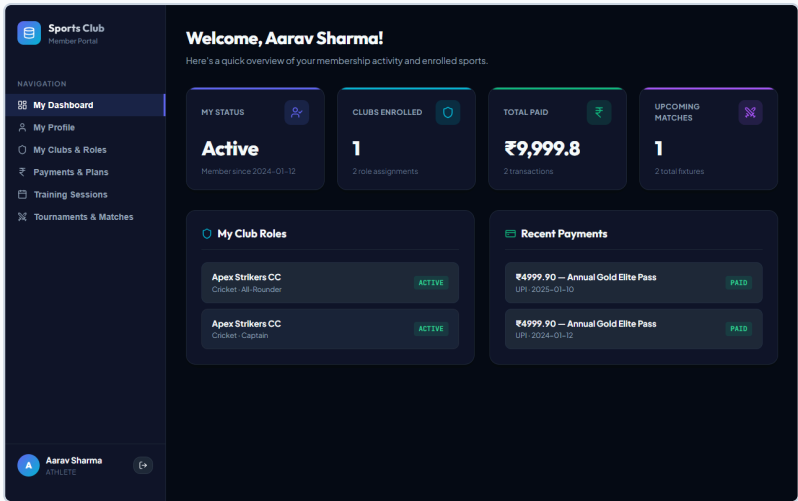


Figure 9.3: Athlete Member Dashboard showing personalized subscription standing and quick club roles.

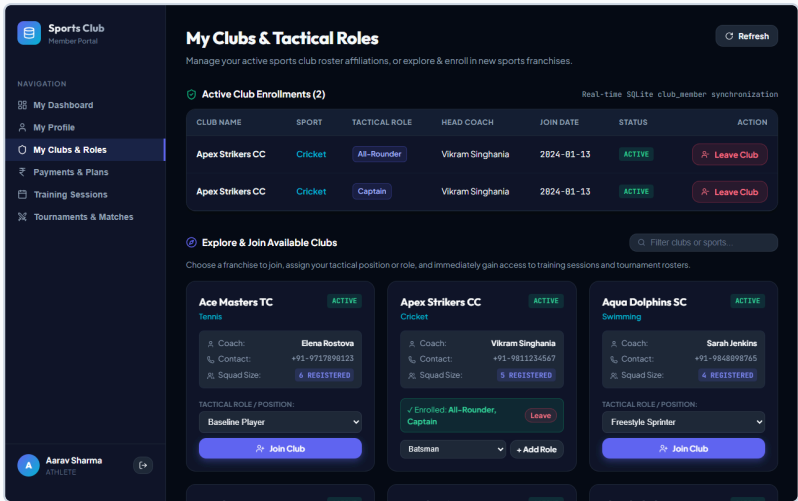


Figure 9.4: Club Management UI displaying active enrollments with "Leave Club" action and "Explore & Join Available Clubs" grid.

9.3 3D Virtual Card Checkout & Official Tax Invoice

The checkout interface simulates real-world payment channels including UPI / QR Code (with 10-minute dynamic expiration), 3D Virtual Credit/Debit Card preview, and Net Banking. Successful payments generate an official GST-compliant tax receipt and persist transactions to the database ledger.

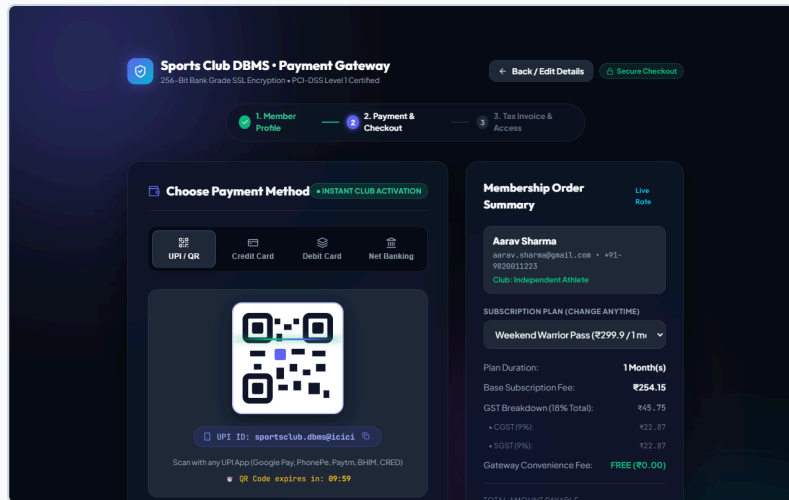


Figure 9.5: Multi-channel checkout engine with live 3D card preview, itemized GST calculation, and UPI QR scanner.

10.0 CONCLUSION & FUTURE SCOPE

10.1 Project Conclusion

The **Sports Club Membership and Tournament Management System** successfully bridges theoretical database principles with practical full-stack software engineering. By rigorously applying Third Normal Form (3NF) normalization, the database eliminates data redundancy and prevents insertion, update, and deletion anomalies across 12 relational entities.

- **Robust Relational Architecture:** Enforced composite primary keys, foreign key cascades, and automated triggers protecting against venue scheduling conflicts.
- **AI-Assisted Database Operations:** Successful integration of the Smart SQL Studio, enabling natural language data exploration without compromising SQL syntax validity.
- **Complete Member Lifecycle:** Dynamic club enrollment, tactical role assignments, realistic payment processing, and automatic cascade updates across training and fixture modules.
- **Verified Integrity & Performance:** Consistent sub-millisecond execution times across complex multi-table joins, window functions, and financial aggregations.

10.2 Future Enhancements

- **Biometric Attendance Integration:** Hardware integration with RFID/biometric turnstiles at sports facilities to verify active membership status automatically upon physical entry.
- **Mobile Native Application:** Developing React Native / Flutter mobile clients for push notifications regarding match fixture reschedules and training session reminders.
- **Predictive Analytics:** Utilizing machine learning models to forecast venue utilization rates and identify athlete churn patterns based on payment history.

11.0 REFERENCES & APPENDIX

11.1 Academic & Technical References

1. Elmasri, R., & Navathe, S. B. (2015). *Fundamentals of Database Systems* (7th ed.). Pearson.
2. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database System Concepts* (7th ed.). McGraw-Hill.
3. SQLite Consortium. (2024). *SQLite Foreign Key Support & ACID Transactions*. <https://www.sqlite.org/>
4. PostgreSQL Global Development Group. (2024). *PostgreSQL 16 Documentation: Schema Design and Triggers*. <https://www.postgresql.org/docs/>

11.2 Project Repository & Source Code Link

Official GitHub Repository
Source code, SQL schemas, dataset scripts, and web applications are publicly hosted at:
<https://github.com/Convoluted-Dense/Sports-Club.git>

11.3 Step-by-Step Setup & Execution Guide

1. Clone the Repository:

```
git clone https://github.com/Convoluted-Dense/Sports-Club.git
cd Sports-Club
```

2. Zero-Config Launch (Python 3 Built-in):

```
python app.py
```

The server initializes `sports_club.db` automatically and launches the application at `http://127.0.0.1:5000`.

3. Default Demo Login Credentials:

- **Administrator:** Username: Admin | Password: Admin
- **Member Athlete:** Email: aarav.sharma@gmail.com | Password: aaravsharma